Developing a Thermoplastic Device Capable of Partitioning Samples for dPCR

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Introduction

- Digital Polyemrase Chain Reaction (dPCR) is a method for the absolute quantification of genetic material
- Upon injection, the sample is partitioned into ~20,000 nanoliter-scale chambers for amplification, and target concentration is estimated via Poisson statistics from the number of fluorescence-positive partitions.
- The aim of this project is to develop a low-cost, reproducible platform for dPCR

Design Objectives

Our objective is to optimize the fabrication process of the polycarbonate dPCR devices, primarily the imprinting and bonding procedures.

Microfluidics Device Design

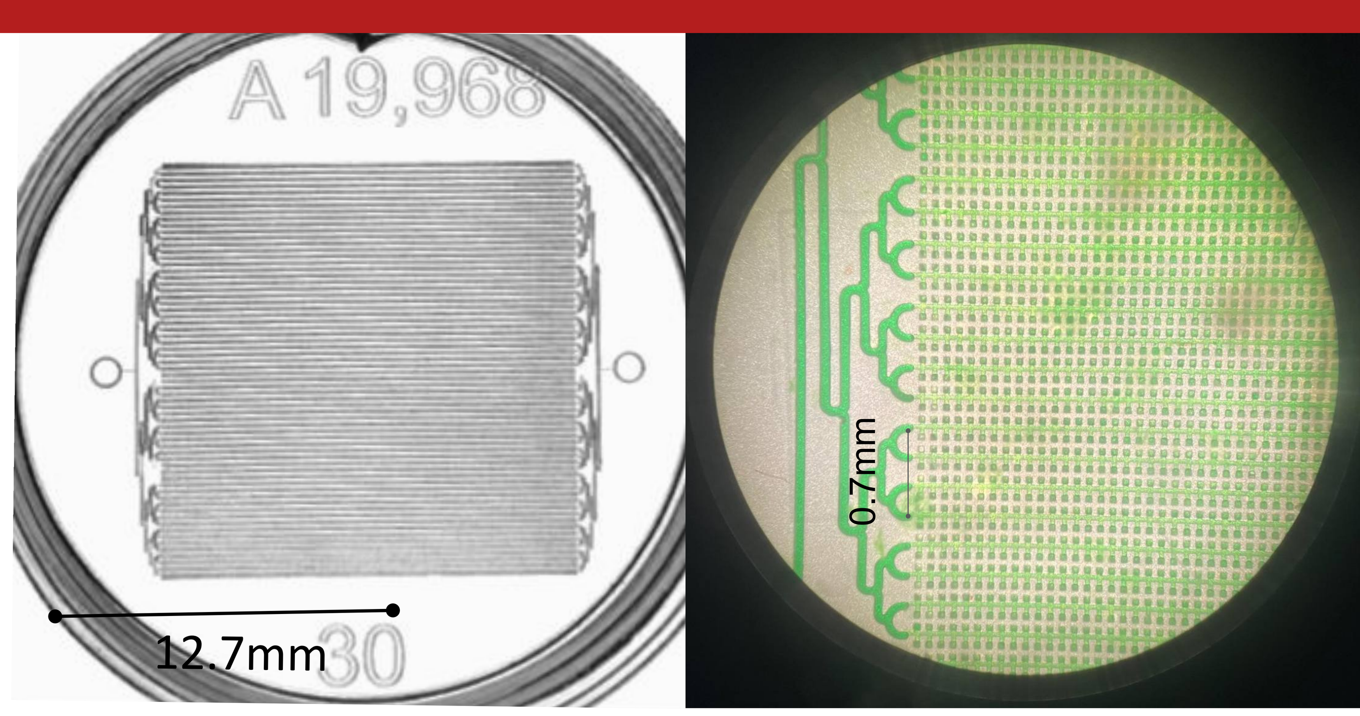

Figure 1: Left: Top down view of the polycarbonate device, showcasing the vast density of chambers and the inlet/outlet ports. Right: Microscope image of microfluidic device showing green dye flow through channel network.

Polycarbonate Device Fabrication

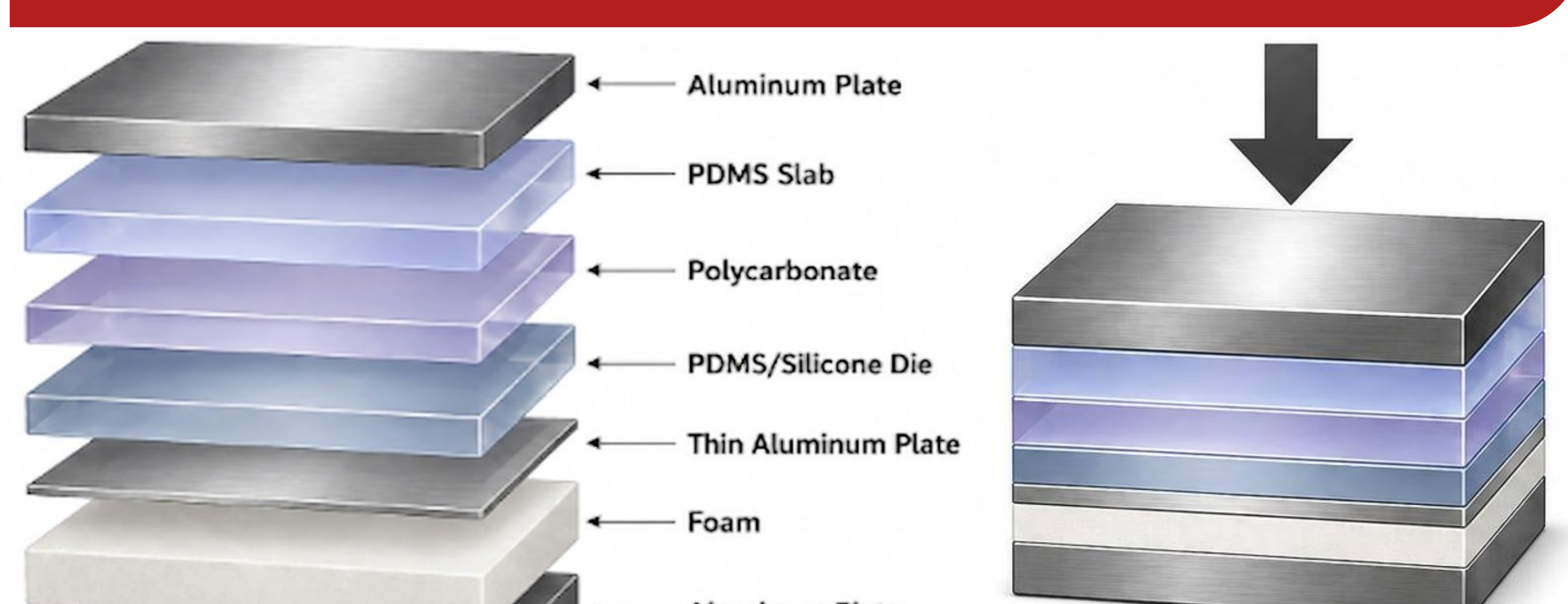

Figure 2: Schematic of the assembly of the apparatus used to imprint the polycarbonate with the design structures

Thermal Bonding

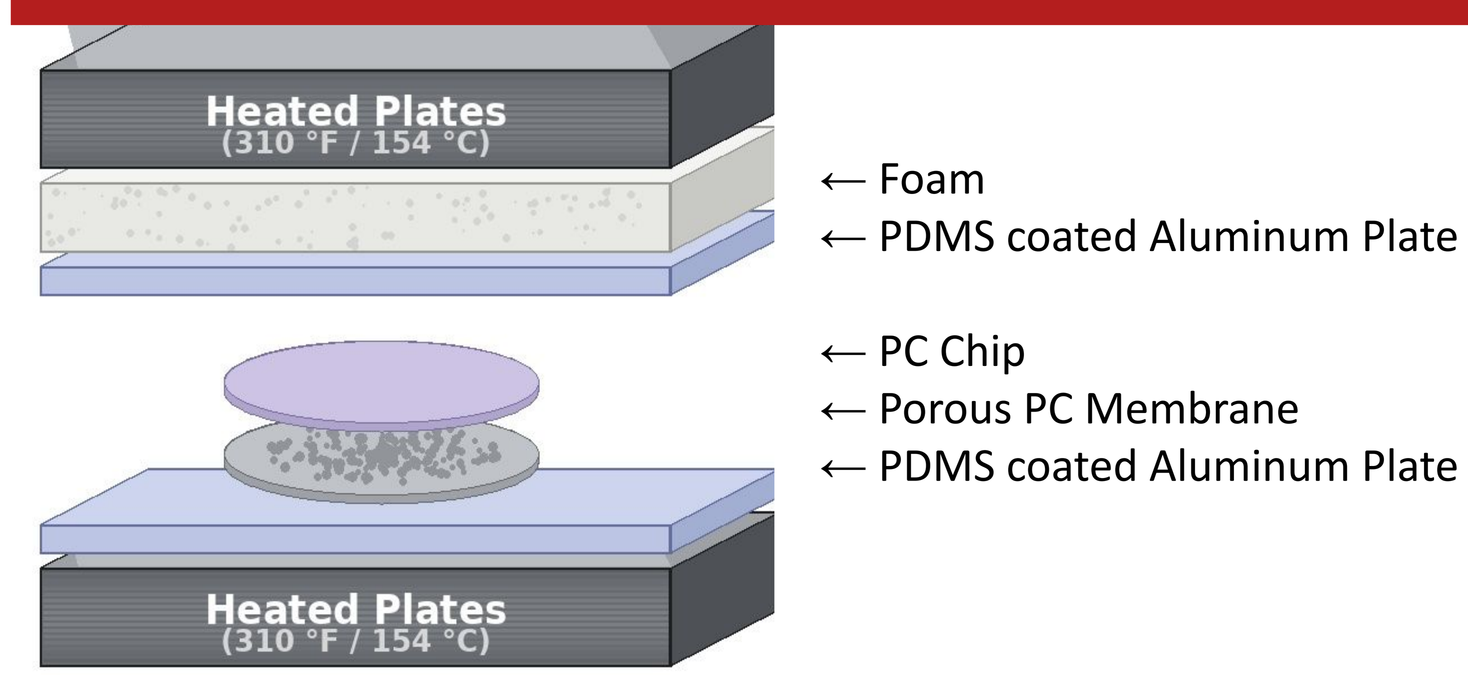

Figure 3: Schematic of the assembly of the apparatus used to thermally bond the membrane to the PC chip.

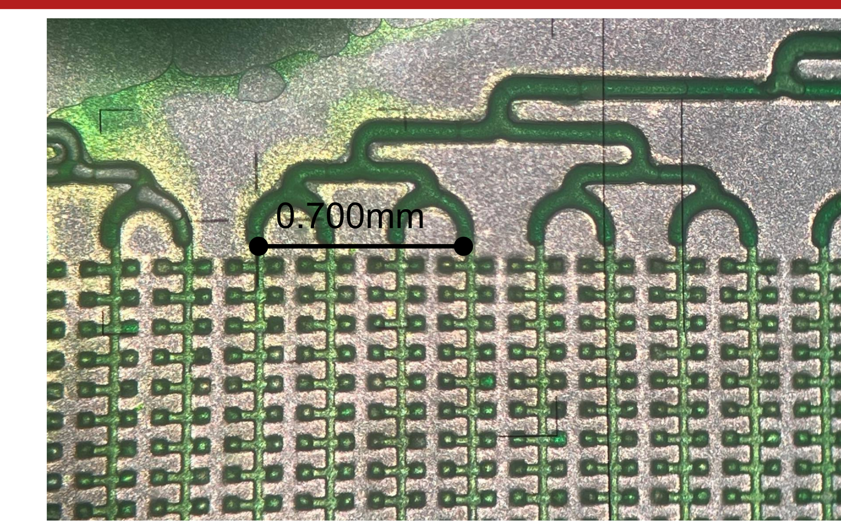

Figure 4: Microscope image of the fully bonded microfluidic device testing with green dye

- Thermal bonding process yielded devices that were capable of aqueous phase testing.
- However, the oil phase were not successful as the technique would often deform the channels.

UV Curable Adhesive Bonding

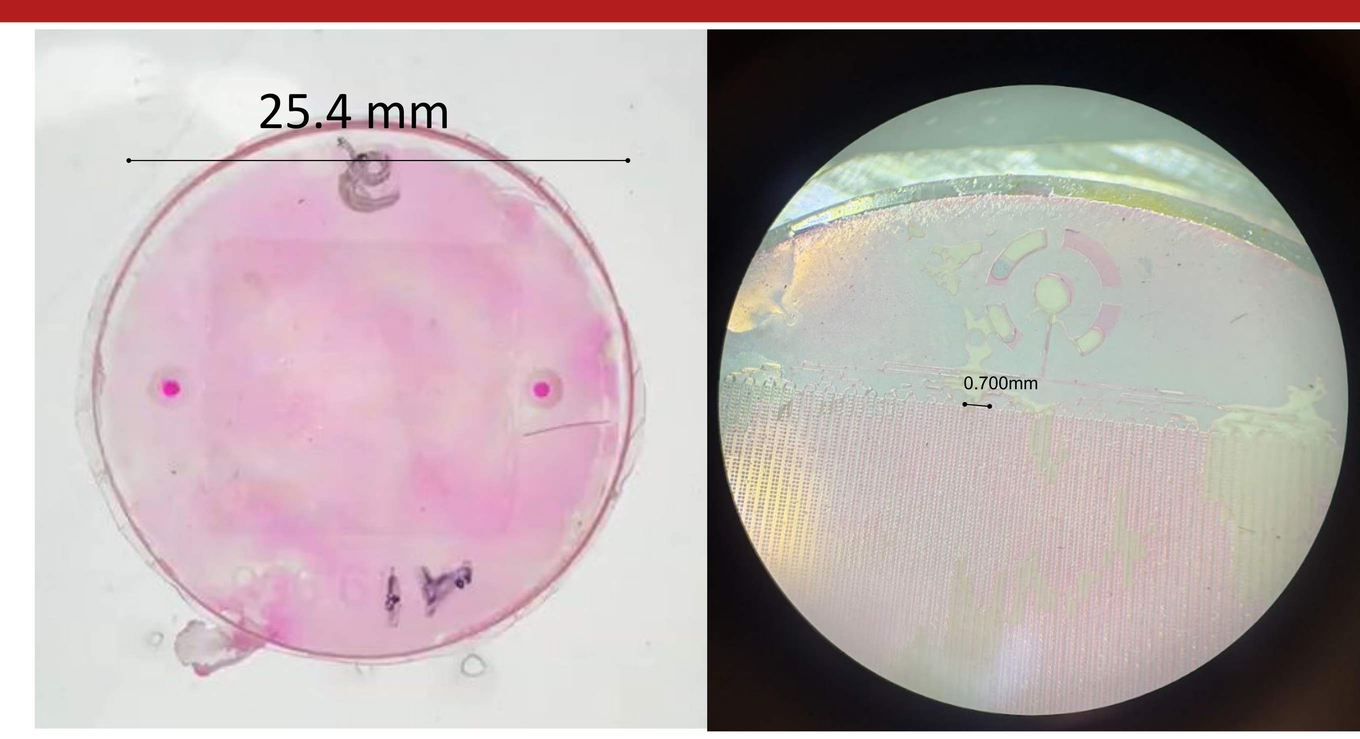

Figure 5: Left: Top down view of the polycarbonate device after Norland Electronic Adhesive 121 Red, followed by UV exposure for bonding. Right: Microscope Image to show adhesion to outer layers of device.

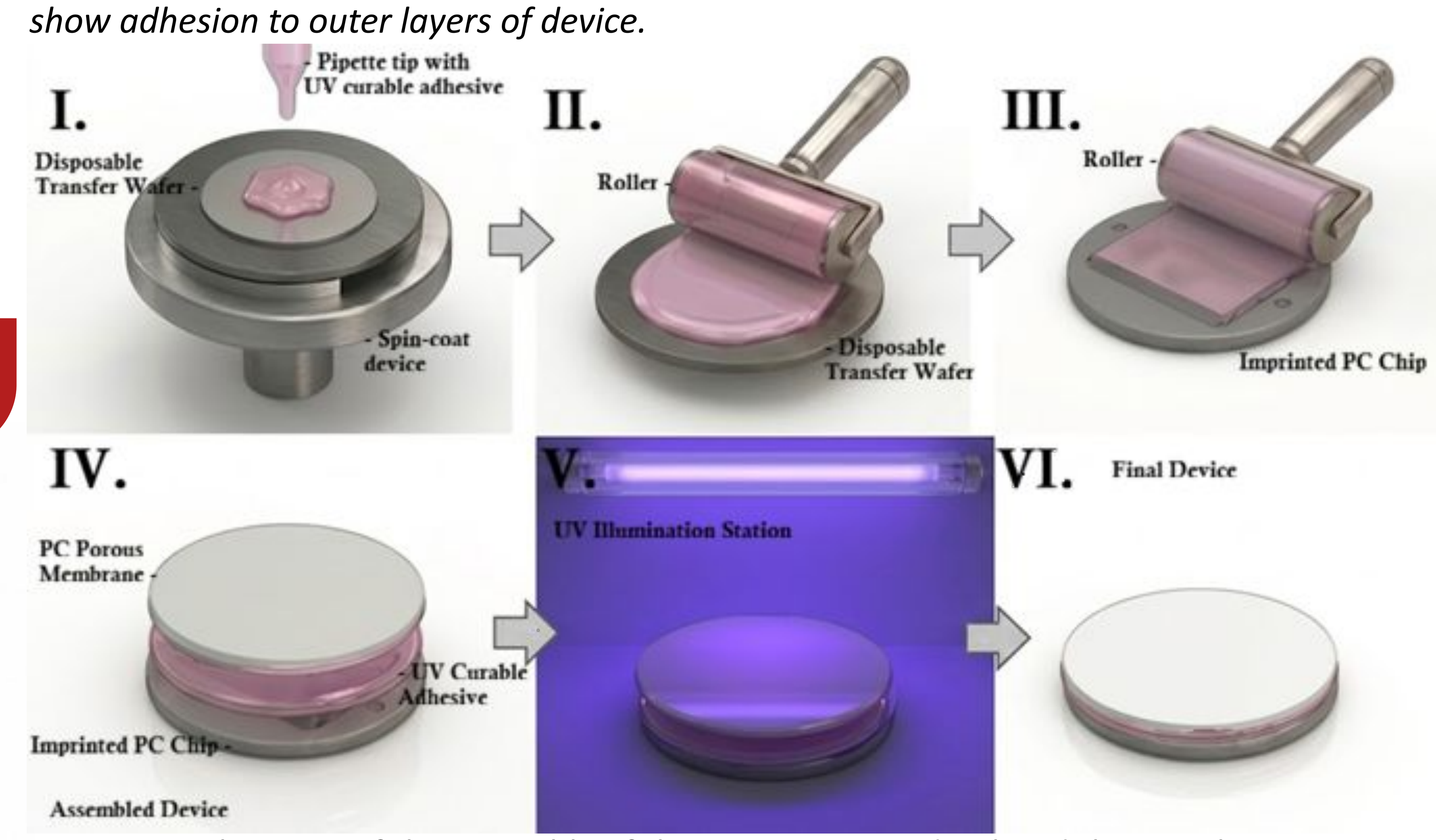

Figure 6: Schematic of the assembly of the apparatus used to bond the membrane to the PC chip using the UV curable adhesive bonding method. (I) Spincoatting the UV curable adhesive onto a transfer wafer for a thinner layer. (II) Picking up adhesive from the wafer using a roller. (III) Applying the adhesive via the roller onto the PC chip. (IV) Assembling device with porous membrane onto chip and adhesive. (V) Incubating assembled device to UV exposure for curing. (VI) Final device ready to use.

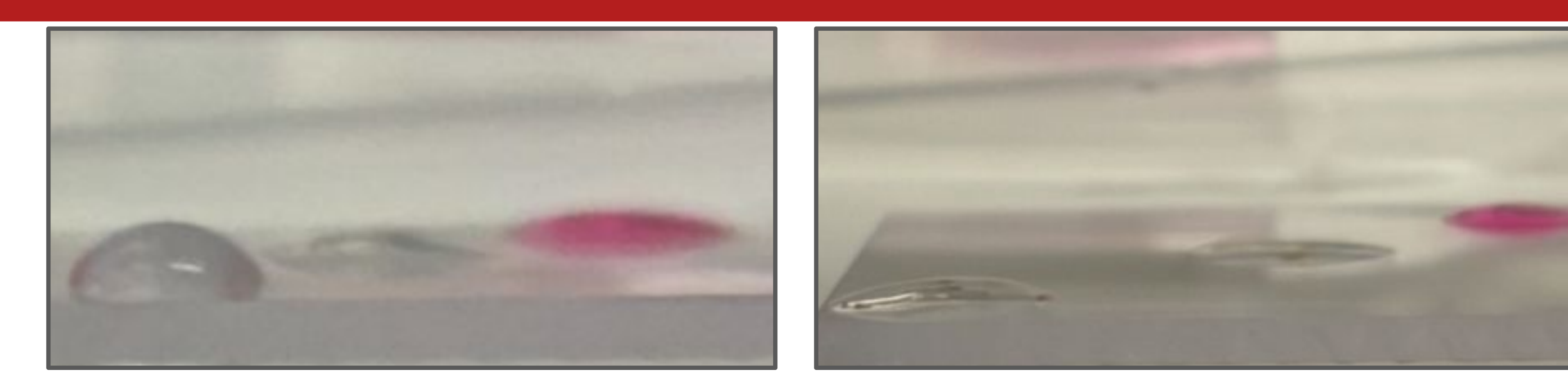

Figure 6: Comparison of polycarbonate material adhesion properties with DI Water, NOA72, and NEA121RED. Left: Polycarbonate with angles 35.2, 23.9, 24.6 degrees respectively. Right: Plasma treated Polycarbonate with angles 10.7, 9.4, 20.2 respectively. Contact angle measurements were taken from side profile and analyzed with ImageJ.

Future Objectives

- Optimize bonding conditions using both the thermal bonding and UV curable adhesive methods
- Proceed to testing with an oil phase, and then proceeding to biological sample testing.

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References

Arayanarakool, R., et al. "Low-Temperature, Simple and Fast Integration Technique of Microfluidic Chips by Using a UV-Curable Adhesive." Lab on a Chip, vol. 10, no. 16, 2010, pp. 2115–2121.